GLA University, Mathura
Department of Computer Engineering and Applications

Assignment (Module I)

Subject: Introduction to Microprocessor
Subject Code: BCSC 0007
Submission deadline: February 20, 2026

Assignment 1.1: Introduction to Microprocessor

1. A team of embedded design engineers has selected 8085 to perform a data processing task. Describe the architecture of the 8085 microprocessors, including its key components and their functions for its successful implementation. Additionally, explain the role of the Program Counter (PC), Stack Pointer (SP), Flag Register (PSW), and various addressing modes in the execution of instructions on the 8085.

2. You are a college student in 2026, preparing for your microprocessor lab exam using the Intel 8085 trainer kit in your college lab. During hands-on practice, you notice that the same data can be accessed in different ways depending on whether it's already in a register, stored at a fixed memory address, or located relative to the stack pointer, and this choice significantly affects both program speed and code size.
   Discuss the different addressing modes supported by the Intel 8085 microprocessor with clear examples showing how each mode is used in typical 8085 assembly programs.

3. Observe and write the contents of the Accumulator and the Processor Status Word (PSW) (in Hexadecimal format) after the execution of the following sequence of instructions. Assume [2000H] = 0AH.

```
STC
MVI     B,  5CH
LXI     D,  2000H
LDAX    D
RAR
ADD     B
HLT
```

4. Distinguish the following instructions of 8085
   (i) SIM and RIM
   (ii) RAL and RLC

(iii)LDAX and STAX
(iv)IN and OUT

5. You are an embedded engineer in for troubleshooting a slow-performing digital weighing scale built around the Intel 8085 microprocessor. The scale frequently reads sensor values from memory and registers for calibration, and finally stores the final weight in memory for display. Compare the following Intel 8085 instructions which will helpful for implementing weighing scale program. Also state the addressing modes, machine cycles, and T-states for each instruction.

   (i) MVI & MOV
   (ii) STA & LDA
   (iii)INX & DCX
   (iv)ADC & SBB
   (v) ANI & XRI
   (vi)RRC & RAR
   (vii) LHLD & SHLD
   (viii) CALL & JUMP
   (ix)INR & DCR
   (x) SIM & RIM
   (xi)LXI H & MOV A, M

6. In a microprocessor lab session, your instructor gives you a task to efficiently swap and manipulate five temperature readings using only the available registers without unnecessary memory access. Describe the complete register array of the Intel 8085 microprocessor, explaining the function and typical use of each register with short examples relevant to handling and averaging sensor readings in your lab assignment.

7. Draw the pin diagram of 8085 microprocessor and explain the functions of DMA, RESET, Status and control pins of 8085.

8. IF [HL] = 2005 H and [A] = 03 H, then what are the contents of location 2005 after execution of MOV M, A and STA 2005 instructions?

9. Write an assembly language program for addition of two 8-bit numbers with a 16- bit sum. The first number is 98 H and is in memory location 2501 H and the second number is 9A H and is in memory location 2502 H. The result is to be stored in 2503 H and 2504 H memory locations.

10. What will be the contents of the status flags after the execution of different compare instructions in the following sequence of Intel 8085 instructions. Assume [A]=6D H and [24FF] = 49 H.

Memory Address	**Hex Codes**	**Mnemonics**	**Operand**	**Comment**
2000 H	21	LXI	H, 24FFH	[H-L]← 24FFH

2001 H	20			
2002 H	01			
2003 H	BE	CMP	M	[A] – [[H] [L]]
2004 H	FE	CPI	6F H	[A] – 6F
2005 H	6F			
2006 H	FE	CPI	6D H	[A] – 6D
2007 H	6D			
2008 H	C3	JMP	2008 H	
2009 H	08			
200A H	20			

11. Discuss the concept of stack in 8085. Also, exchange the contents of BC and DE register pairs using stack instructions only.

12. Load the hexadecimal numbers 9BH and A7H in registers D and E, respectively, and add the numbers. If the sum is greater than FFH, store 01H at memory location 24FF H otherwise store the sum at memory location 2500H.

13. What do you mean by instruction word size? Explain using suitable example.

14. Write assembly language programs for the following

    (i) Pack the two unpacked BCD numbers stored in memory locations 3200H and 3201H and store result in memory location 3300H. Assume the least significant digit is stored at 3200H.

    (ii) Two-digit BCD number is stored in memory location 4200H. Unpack the BCD number and store the two digits in memory locations 3300H and 3301H such that memory location 3300H will have lower BCD digit.

15. Consider you're tasked with designing a timing diagram for the 8085 micro-processor to illustrate the execution of following instructions:

    (i) LHLD 2FFFH
    (ii) CPI 08H
    (iii) LDA 3201 H
    (iv) LXI H, 2500H
    (v) INR M
    (vi) MVI M, 07H

    Sketch the timing diagram showing the sequence of events and their timings for each instruction.

16. Discuss the instruction cycle, machine cycle and T-state.

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Assignment 1.2: Introduction to Microprocessor

1. A student has just started learning 8085 microprocessor programming and faces difficulties while writing assembly programs. Describe the challenges faced by the student and explain how these challenges can be overcome.

2. An engineer designing an 8085-based system observes that the address bus and data bus share the same pins. Explain why multiplexing is used and analyze its advantages.

3. A programmer wants to evaluate execution time for instructions such as ADD B, STA 2005H, IN 43H, and MVI A, 58H. Determine the number of machine cycles and T-states required.

4. While explaining the working of the 8085 microprocessor, an instructor asks students to visualize its internal structure. Draw the internal architecture of the 8085 and explain the functions of registers, ALU, stack pointer, and program counter.

5. An 8085 microprocessor operates at 2 MHz and executes four NOP instructions to generate delay. Calculate the total time delay produced.

6. During an arithmetic operation, two numbers 07H and CFH are added. Explain the role of the Program Status Word and analyze the status of all flags after execution.

7. An application uses rotate instructions on accumulator data where A = B7H and CY = 0. Find the accumulator content after executing RLC twice and RAL twice.

8. A system stores numbers in unpacked format and requires them to be packed before further processing. Write an 8085 assembly language program to pack two unpacked numbers.

9. While learning operand access methods, a student studies addressing modes of the 8085. Explain any three addressing modes with two examples each.

10. In an 8085 system running at 3 MHz, machine codes 3EH and 32H are stored in memory. Calculate the time required for opcode fetch, memory read, and the entire instruction cycle.

11. During hardware assembly, improper power connections cause malfunction. Explain the importance of Vcc and Vss pins and discuss possible issues if they are not connected properly.

12. The accumulator contains AAH and the carry flag is reset. The RAL instruction is repeatedly executed until the accumulator becomes A9H. Determine the number of executions and analyze the carry flag status.

13. A program needs to add data from memory locations 3000H and 3001H and store the result at 3002H. Write an 8085 assembly language program to perform this task.

14. A user sends a text message using WhatsApp. Justify whether WhatsApp is a program or data and explain the process in terms of instruction cycle (Fetch, Decode, Execute).

15. An 8085 executes the instructions LXI H, 25A0H; DAD H; and PCHL. Determine the contents of the Program Counter and HL register pair after executing PCHL.

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Assignment 1.3: Introduction to Microprocessor

1. Suppose your job interview is scheduled, and the interviewer asks you some questions based on the 8085 microprocessor. You have to answer them by just stating the name of the instruction.
   (a) Write the name of the instruction which can store the program counter address into the stack?
   (b) Write the name of the instruction in which the content of accumulator is rotated right by one bit. The LSB bit of accumulator moved into the MSB as well as in carry flag?
   (c) Write the name of the instruction which is used to increment the data of memory location pointed by HL pair?

2. Imagine you have a small storage box (memory location 4200H) that contains a two-digit number written together, like "47". Your task is to carefully separate the two digits and place them into two different labelled drawers (memory locations) with the help of assembly language program.

   The ones place (7) should go into drawer 4300H.

   The tens place (4) should go into drawer 4301H.

3. What is the role of DAA instruction in our daily life? Define it with suitable example.

4. What is the reason behind that the address is given just next to Call instruction and address is not given just next to RET instruction?

5. Imagine you are working as a software engineer at a tech company that designs embedded systems. Your team is assigned for developing a small embedded system that uses the 8085 microprocessor to perform basic arithmetical/logical operations. You have been asked to explain the working of 8085 microprocessor architecture with the help of an instruction. In which explain that how the 8085 microprocessor will fetch, decode and execute an instruction? Also provide a neat and detailed diagram of the 8085 microprocessor architecture to illustrate its functioning.

6. As an interviewer I want to ask:
   A. Write the name of the instruction which can reset the lower nibble without affecting the higher nibble of the data of Accumulator?

   B. Write the name of the instruction which can set the higher nibble without affecting the lower nibble of the data of Accumulator?

   C. Write the name of the instruction which can complement the lower nibble without affecting the higher nibble of the data of Accumulator?

7. What will happen if DMA controller will not be available for microprocessor world? How it will affect our daily life? Which pins are used for its function in 8085 Microprocessor? Define its pins.

8. Why the sign flag is not a faithful flag in the 8085 Microprocessor? Which flag can overcome the issue of sign flag and how?

9. Whenever we perform the CMP B instruction for compare so there are three possibilities A>B, A=B and A<B. Which flags will be affected by A>B, A=B and A<B. Write the name of the flags with suitable status for each above three conditions.

10. How the call instruction defines the operation of interrupts? What do you mean by interrupt service routine program?

11. How the architecture of the 8085 microprocessor operates, including its ALU, Registers and Flags, and main components that interact with the data bus and address bus. Draw its suitable diagram.

12. Why the interrupts play the major role in microprocessors world? How does an interrupt mechanism work in the 8085 microprocessor? Discuss the different types of interrupts and their priority levels.

13. The 8085 microprocessor operates at a particular clock frequency. Please explain how this frequency is generated and discuss its importance. What is need of clock in microprocessor world?

14. Compare and contrast the features and capabilities of the 8085 microprocessor with other popular microprocessor from the same era.

15. In the context of the 8085 microprocessor, what is the purpose of the stack, and how does it operate with respect to push and pop operations?
16. Define, How the function of interrupts can be explain with the help of CALL 3500H and RET instructions?